

CULTIVARS

- Savor (charentais)
- Escorial (charentais)
- Halona (cantaloupe)
- Athena (cantaloupe)
- Sugar baby (watermelon)

INTENSIVE SPACING

- 1 row per bed every 30" (75 cm) on the row

FERTILIZATION

- Melons do well in beds amended with lots of compost, fresh or aged. The rule of thumb is to spread 1 inch (5 cm) in the middle of the bed. Since it's a single row crop, there's no need to apply the compost on the whole width of the bed.
- When planting in beds where compost was applied in the previous crops, we only supplement with a mixture of alfalfa meal (2-0-2) and pelleted chicken manure (5-3-2).

IRRIGATION

Irrigation Type

- Drip tape (4 per bed) : under plastic mulch.
- No sprinkler to keep foliage dry and thus prevent diseases.

Irrigation Management

- Root depth: Medium (24"-40")
- General: Rapid growth and ongoing fruit production requires frequent deep water. Critical to water consistently at flowering and fruit development stage.

STARTING SEEDS

- 2 weeks before transplantation date, sow 1 seed per cells in 72 plug flats. Seeds should be about 1/4" (5mm) deep in the cell.
- Place trays on heating mats and maintain soil temperature at 86 °F (30 °C) until emergence. At that time, the potting-mix must be kept moist at all times.
- After emergence, the ambient temperature in the nursery can at set be kept around 75°F (24 °C) during the day and 65°F (18 °C) at night.
- 4 days before transplanting, harden off plants by exposing them to outside temperatures and reducing water.

IMPLANTATION

We plan for only 1 melon seeding of both charentais, cantaloupe and watermelon. For all of these, we stagger our succession (to avoid bumper crops) by planting 2 varieties with different DTM one early followed by another one better tasting.

EQUIPMENT

- Rotary harrow
- Broadfork
- 30" bed prep rake
- Declumping tray
- Landscape fabric, staples, drip tape and fittings.

TRANSPLANT ITINERARY

- Pass through with the broadfork.
- Remove large debris from the bed if needed (use flail mower, rake or manually remove the large chunks).
- Apply compost and-or amendments in the middle of the bed and pass with the rotary harrow at a depth of 2.5" (6 cm). Since the bed is covered with landscape fabric, it needs not to be clean up with another pass of the rake.
- Install 4 lines of drip tape per bed and start the system. Fix leaks at that point, if any.
- Lay down landscape fabric and secured with staples each second step (5' or 1.5 m). It needs to be really tight.

- Transplant melons making sure to place them in the center of each hole. Melon seedlings are root-sensitive. Handle them carefully and don't bury their collars.
- Water the bed thoroughly the first time, keeping the soil moist for the first week after the transplant.
- Cover the crop with insect netting (late spring) or floating row covers (early spring).

WEEDING

- No weeding required

PLANT PROTECTION

- Young melon plants must always be covered with an anti-insect net (or floating row cover in the spring) to prevent both the cucumber beetles and squash beetles from establishing themselves.
- When the plants begin to flower, the net must be removed. At that point, the plant is well established and should survive the presence of both the cucumber and squash beetle.
- Melons also susceptible to plant diseases such as mildew and fusarium wilt. Disease-resistant varieties is the first line of defence. Growing melons under shelter is the second. Taking out all infected plants and destroying them is important.

MAINTENANCE

- When the plant has 4 true leaves, prune off the main stem that will then branch off with 2 heads (instead of one). This helps the plant bear more fruits.
- During peak flowering (+/- 6 weeks after transplanting), spray plants with a foliar application of seaweeds (which contains minerals and micronutrients). This helps the plant develop as much as possible before it starts to bare fruits.
- When melons start to reach the size of a baseball, drastically reduce all irrigation to about 1/3 of previous amounts. This will increase the sugar in the fruits.
- At that time, (+/- 9 weeks after transplanting) heavily prune the plants in the pathways. This will allow you to circulate better for the harvest. It also stresses the plant to go into fruiting mode. Do this operation only on sunny days to avoid the possible spread of plant diseases.

HARVEST

→ Harvest only begins when fruits are perfectly ripe (if too early, the sugar production will be stunted and, if too late, the melon will have a fermented taste and be soft)

Melons will have different indicators of ripeness

→ **Charentais:** when they reach their size and go from a light green to yellow coating. Slice into a melon to test for ripeness.

→ **Cantaloupe:** when they reach their size and the peduncle detaches itself from the fruit. You can also use your nose: when it smells like a delicious ripe melon, it's ready to be harvested. Slice into a melon to test for ripeness.

→ **Watermelon:** when they reach their size and the underside of the melon yellows and the tendril closest to the fruit starts to wilt. A ripe watermelon should make a hollow sound when you tap it. Slice into a melon to test for ripeness.

→ Harvest in tree planting bags and cut the fruits using a small pruner. Once the bags are filled, gently deposit the melons in harvest containers.

→ Come every second day to make sure you harvest each melon at the right time.

CONDITIONING

→ Melons are not washed but must be wiped clean if dirty.

STORING

→ Melons are stored in cold room kept at 47 °F (8 °C) and should not be stored for more than 4 days before being sold.